

Observing Case: Does the Galactic Acceleration Scale Evolve? One Deep-MOND Rotator at $z \simeq 2.5$ Decides

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AI-assisted research programme; not peer reviewed

11 September 2026

Abstract

Companion to the pre-registered measurement (doi:10.5281/zenodo.22563139), written for observers. Two readings of the radial-acceleration scale $a_0 \simeq 1.2 \times 10^{-10} \text{ m s}^{-2}$ predict opposite redshift behaviour: tied to the dark-energy density, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, it is constant; as the emergent scale of Λ CDM haloes it tracks $H(z)$ and rises by $2.1^{1/4} = 1.20$ in rotation speed at fixed baryonic mass by $z \simeq 2.5$. The frozen statistic is the deep-MOND Tully–Fisher zero-point shift, 0.00 dex (constant) versus +0.33 dex (rising), decidable at ± 0.13 dex with a single galaxy that is genuinely deep-MOND, rotation-dominated, gas-measured and lens-controlled. We give the target gate, the NIRSpec-then-ALMA funnel, the per-galaxy error budget, the current candidate ledger (no known object passes the gate today; the decisive galaxy must be found from low-mass highly magnified lensed discs), the controls, an order-of-magnitude time request, and the frozen decision rule. Either outcome is a publishable, pre-registered result.

1 The two predictions

at $z \simeq 2.5$, fixed baryonic mass	constant a_0	rising $a_0 (\propto H)$
$\Delta_{\text{BTFR}} = \log_{10} M_b - 4 \log_{10} V_f - C_0$	0.00 dex	+0.33 dex
flat speed of a 60 km s^{-1} dwarf	60	72 km s^{-1}
required precision per decisive galaxy	$\sigma(\Delta_{\text{BTFR}}) \leq 0.13 \text{ dex}$	

C_0 is frozen from the local calibration on both a_0 footings (9.36×10^{-11} , $1.13 \times 10^{-10} \text{ m s}^{-2}$) before any high-redshift value is examined. Estimator: galaxy median, then median across galaxies; no drift nuisance is introduced after observation. The existing archive (twelve published constraints) is exhausted and prior-dominated; it cannot decide.

2 The target gate

A galaxy enters the decisive sample only if all hold: (1) $g_{\text{bar}}(R_{\text{out}}) < 0.30 a_0$ on both footings at the upper end of the mass uncertainty, from the resolved stellar and gas distribution (preferred < 0.20); (2) $V_{\text{rot}}/\sigma > 1.5$ from a forward-modelled source-plane disc fit with dispersion, inclination, beam smearing, asymmetric drift, clumps and outflows fitted together (preferred > 2), regular monotonic

field; (3) $\delta \log_{10} \mu < 0.08$ dex with no credible alternative reconstruction outside the budget; (4) $35^\circ < i < 75^\circ$ and ≥ 5 source-plane resolution elements along the major axis; (5) $2.32 < z < 3.12$ so that H α and [O III] fall together in NIRSspec G235H/F170LP and CO(3–2) falls in ALMA Band 3. Failing (1) or (2) means rejection without interpretation.

3 The funnel and the budget

Stage 1, NIRSspec IFU G235H/F170LP: a two-dimensional velocity field and a forward-modelled source-plane velocity cube (never a de-lensed rotation curve); advance only if the rotation gate passes and $g_{\text{bar},*} + g_{\text{gas,max}} < 0.3 a_0$ with a conservative upper gas estimate. **Stage 2, ALMA Band 3 CO(3–2):** integrated H₂ mass and a resolved CO velocity field; two independent tracers are required and a material discrepancy invalidates the target. $M_b = M_\star + 1.36(M_{\text{HI}} + M_{\text{H}_2})$ with X_{CO} and excitation propagated. **Budget:** $\delta V/V < 5\%$ (combined tracers), $\delta \log_{10} M_b < 0.10$ dex, $0.05\text{--}0.07$ dex for inclination, lens model and asymmetric drift, hence $\delta \log_{10} \mu \lesssim 0.05\text{--}0.08$ dex (this alone removes MACS J0451, whose lens models disagree, 21.5 vs 49).

4 Candidates, controls, time

The committed ledger of 21 objects, scored on both footings, contains **no object that passes the flagship gate today**: every measured rotator is high-acceleration and every deep-MOND candidate lacks resolved rotation. Inside the joint window: Cl 0949+5153 arc ($z = 2.39$, merger, rotation unknown); A1689B11 ($z = 2.54$, $V/\sigma = 8.7$, high-acceleration: a control); A68-HLS115 ($z = 2.49$, no kinematics); Abell 68 C4 ($z = 2.622$) and C20b ($z = 2.689$), unscored low-mass lensed candidates with H α , [O III] and CO(3–2) all reachable: the first Stage-1 targets. Outside the window, SL2S 0217 ($z = 1.84$, $\mu = 17$, deep-MOND score 0.92, rotation unknown, CO(4–3) in Band 4) is the best deep-MOND candidate if the window is relaxed to H α only. Mandatory controls: a high-acceleration rotator in the window (A1689B11) to recover the local zero-point where both predictions agree, and a cold-tracer control (zC-400569, $z = 2.24$, $V/\sigma = 20$). Time, order of magnitude and not an exposure-time-calculator run: NIRSspec IFU $\sim 3\text{--}5$ h per candidate (10–20 h for 2–4), ALMA Band 3 $\sim 5\text{--}10$ h per source (10–20 h for 1–2).

5 Decision rule

Near 0.00 dex the constant law survives and the rising reading is excluded at the stated precision; near +0.33 dex the constancy prediction is falsified; inconsistent with both is evidence against both and is reported as such. One galaxy is enough if and only if it passes the gate. A null on the gate itself (no galaxy passes) is informative about the deep-MOND population at $z \simeq 2.5$. Provenance: doi:10.5281/zenodo.22563139; repository `prep_2026/a0z_crossscale/` (ledger, scores, likelihood).